Exponential distribution + inverse exponential

1 RECOGNISE IT

- Waiting time/distance between random events occurring at average rate lambda.
- Right-skewed, support x>0.
- Rate and time units must match.

2 DECISION / METHOD

1. Convert rate and time to the same unit.
2. Left wait probability uses 1-exp(-lambda*t).
3. Right wait probability uses exp(-lambda*t).
4. For a percentile, solve the CDF for t.

3 FORMULA

```
f(x)=lambda*exp(-lambda*x)
F(t)=1-exp(-lambda*t)
P(X>t)=exp(-lambda*t)
t_q=-LN(1-q)/lambda
E(X)=1/lambda; Var=1/lambda^2
```

4 FORMULA SHEET VS WHAT I KNOW

Gives: density, mean and variance.
Still know: left/right CDFs, inverse formula, memoryless context, and unit conversion.

5 EXCEL ROUTE

- Left: `=EXPON.DIST(t,lambda,TRUE)`
- Right: `=1-EXPON.DIST(t,lambda,TRUE)`
- Inverse: `=-LN(1-q)/lambda`.

6 TINY WORKED EXAMPLE

At lambda=4 per hour, `P(wait>0.25h)=EXP(-4*0.25)=0.3679`.

7 TOP EXAM TRAPS

- Mixing minutes with a per-hour rate.
- Using 1-exp(...) for a right tail.
- Treating lambda as the mean wait; mean wait is 1/lambda.

8 SAY ALOUD

Exponential waiting-time questions live or die on matching units and choosing left versus right tail.

9 EXIT CHECK

If F(t)=0.80, what equation gives the 80th-percentile wait?

ONE-GLANCE VISUAL ROUTE

